CENTRE FOR DEVELOPMENT OF ADVANCE COMPUTING

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POST GRADUATE DIPLOMA IN ADVANCE COMPUTING

Project Report on
E-HEALTHCARE PORTAL

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We are very glad to mention the name of Mr. Shakir H. for his valuable guidance to work on this project.

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ABSTRACT

E-Healthcare Portal is a centralized website to facilitate various services to patients and this website can be accessible to Admin, Patients, and Doctors where all services will be placed under a single umbrella. The user is authenticated as they login and based on their roles they are provided with different functionalities.

The system consists of three users namely Patient, Doctor, and Admin, each one of them having their own set of functionalities. Admin performs registrations and removal of Departments and Doctors. Doctors are assigned to Department where they treat patients according to provided schedules and provide them with prescriptions. The Patient can book appointments for various doctors available from various departments and can consult them online (after this they will be directed to the payment gateway to make the payment) or offline. Users of a particular role cannot access other users' functionalities.

1. INTRODUCTION

Health of citizen is the wealth of Nation. India contributes the most ancient medical science "AYURVEDA" to the world besides other medical sciences. This field had witnessed a rapid metamorphosis in all its sections. **E-Healthcare Portal** is designed to improve the quality and management of hospital in the areas of clinical process analysis and activity-based costing. **E-Healthcare Portal** enables you to develop your organization and improve its effectiveness and quality of work. Managing the key processes efficiently is critical to the success of the hospital, helps you manage your processes.

The **E-Healthcare Portal** can be entered using a username and password. It is accessible by an Admin, Doctor & Patient. Only they can add data into the database. The data can be retrieved easily. The data is well protected for personal use and makes the data processing faster**. E-Healthcare Portal** is powerful, flexible, and easy to use and is designed and developed to deliver real conceivable benefits to Patients.

The project **E-Healthcare Portal** includes registration of patients, storing their details into the system by using database. The software has the facility to give a unique id for every patient and stores the details of every patient and Doctor. Admin can add/remove Department, Doctor and can see the details of the Patient.

2. PRODUCT OVERVIEW AND SUMMARY

- **PURPOSE:**

Our project, "**E-Healthcare Portal**" is a web-based online appointment booking application which aims to provide patient with an easy to navigate and visually appealing medium to browse and book their appointments within a respective department with a doctor of their choice.

It enables doctors and admin to view and modify appointments schedules if required. The purpose of this project is to computerize all details regarding patient and doctor.

- **SCOPE:**

"**E-Healthcare Portal**" aims to deliver a web-based application that helps to increase the number of patients that can be treated and managed properly. Patients can book appointment, see their previous prescriptions, and give their feedback for each doctor. They can update their profile and can pay the fees using an online payment gateway. They can view their previous medical records as well. Admins can manage various departments, doctors and can view patient records. This portal is only an interface between patient and doctor for their ease in the treatment process.

- **OVERVIEW:**

- ## TECHNOLOGIES USED

i. FRONT END

- HTML
- CSS
- JavaScript
- Bootstrap
- React
- Axios

ii. BACK END

- Spring Boot
- Spring Data JPA
- Hibernate
- REST

iii. DATABASE MANAGEMENT SYSTEM

- MySQL

➢ FEATURES PROVIDED:

- JWT (Java Web Token) Spring Security
- Image Handling
- AOP (Aspect Oriented Programming)
- RazorPay API
- SMTP (Simple Mail Transfer Protocol)
- Cloud Deployment

➢ FUNCTIONALITY PROVIDED:

For Admin:

- **Login as Admin:-** Authenticated Admin details are fetched.
- **Add Doctor:-** New Doctor details are added in the database
- **Add Department:-** New Department is added in the database
- **Delete Doctor:-** Associated Doctor details are deleted from the database
- **Delete Department:-** Department is deleted from the database
- **Delete Patient:-** Patient details are deleted from the database
- **Update Doctor:-** Updated doctor details are saved in database
- **Get Department List:-** Department list is fetched from the database
- **Get Doctor List by Doctor Name:-**Got the doctor by searching his name
- **Logout:-** The session was cleared

For Doctor:

- **Login as Doctor:-** Authenticated Doctor details are fetched.
- **Show today Patient List:-** Today's appointments of patient are fetched from the database.
- **Show future Patient List:-** Future's appointments of patient are fetched from the database.
- **Show Patient's Appointment Request:-** Pending appointment list of all the patient is fetched.
- **Get Patient Details:-** All the details of particular patient were fetched.
- **Cancel appointment:-** Appointment was cancelled by doctor and also its status is updated from "Pending" to "Cancelled" into database.
- **Add Prescription:-** All the prescription given by the doctor to patient is saved in database.
- **Get Patient History:-** All the records of patient were fetched from the database.
- **List of Lab Test:-** All the lab tests entries were fetched from the database.
- **Logout:-** The session was cleared.

For Patients:

- **Register as Patient:-** New Patient details saved in the database
- **Login as Patient:-** Fetched authenticated patient details
- **Book Appointment:-** New appointment of a doctor is booked successfully
- **Get Patients Appointment:-** All the appointments booked by patient are fetched
- **Cancel Appointment:-** Appointment is cancelled by patient
- **Get Patient Treatment Details:-** All the treatment details of patient done till now are fetched.
- **Update Patient Details:-** Patient updated his profile.
- **Feedback of Doctor:-** The new feedback associated with the doctor was saved in the database.
- **Online Payment:-** Online payment is done by patient via Razor Pay
- **Logout:-** The session was cleared

3. REQUIREMENTS FULFILLED

3.1 Functional Requirements:

Following are the functional requirements fulfilled by our project:

- Patients can view the scheduled appointments, book a new appointment, or cancel the booked appointment.
- Patient can be able to edit his profile and can update his photo.
- Patient can view the treatment history and the tests prescribed to him.
- Patients can also give feedback to the doctor by whom the patient got the treatment.
- Patient can pay doctors' fees online.
- Doctors can View and cancel today's, Future, and pending appointments.
- Doctors can view patient details, history of patients and also can be able to give prescriptions to the patient.
- Admin can add and remove the departments
- Admin also has the facility to add, update remove certain doctor's information.
- Admin can view the patient details and can delete the patient.

3.2 Non-Functional Requirements:

Following are the non-functional requirements fulfilled by our project:

- Since the application used lightweight and established software components that are also cross-platform, it is remarkably performant and good support for every operating system.
- The use of React for front end and Spring Boot, and Spring Data JPA for the back end delivers quick response time to admins, doctors, and patients alike.
- Card-style UI used throughout the application provides a consistent theme and user-friendly interface that anyone can grasp easily, even without a technical background.

4. PROJECT DESIGN

4.1. DATA MODEL:

The following tables depict the database design used for “E-HealthCare” application:

1. **Department Table:**

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
id	bigint	NO	PRI	NULL	auto_increment
description	varchar(255)	YES		NULL	
department_name	varchar(20)	YES		NULL	

2. **Doctor Table:**

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
id	bigint	NO	PRI	NULL	auto_increment
email	varchar(50)	YES	UNI	NULL	
gender	varchar(10)	YES		NULL	
name	varchar(30)	NO		NULL	
password	varchar(100)	YES		NULL	
education	varchar(255)	YES		NULL	
fees	double	NO		NULL	
is_validated	bit(1)	NO		NULL	
is_working	bit(1)	NO		NULL	
speciality	varchar(255)	YES		NULL	
departmen...	bigint	NO	MUL	NULL	
image_path	varchar(255)	YES		NULL	

3. **Admin Table:**

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
id	bigint	NO	PRI	NULL	auto_increment
email	varchar(50)	YES	UNI	NULL	
gender	varchar(10)	YES		NULL	
name	varchar(30)	NO		NULL	
password	varchar(100)	YES		NULL	
image_path	varchar(255)	YES		NULL	

4. **Lab Test Table:**

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

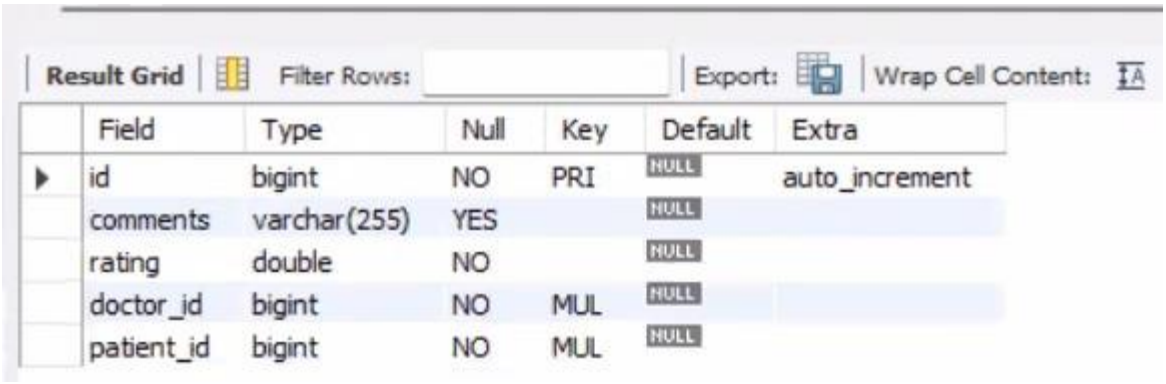
Field	Type	Null	Key	Default	Extra
id	bigint	NO	PRI	NULL	auto_increment
test_name	varchar(20)	YES		NULL	

5. **Patient Table:**

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
id	bigint	NO	PRI	NULL	auto_increment
email	varchar(50)	YES	UNI	NULL	
gender	varchar(10)	YES		NULL	
name	varchar(30)	NO		NULL	
password	varchar(100)	YES		NULL	
address	varchar(100)	YES		NULL	
blood_group	varchar(255)	YES		NULL	
dob	date	NO		NULL	
height	double	NO		NULL	
weight	double	NO		NULL	
image_path	varchar(255)	YES		NULL	

6. **Feedback Table:**



Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
id	bigint	NO	PRI	NULL	auto_increment
comments	varchar(255)	YES		NULL	
rating	double	NO		NULL	
doctor_id	bigint	NO	MUL	NULL	
patient_id	bigint	NO	MUL	NULL	

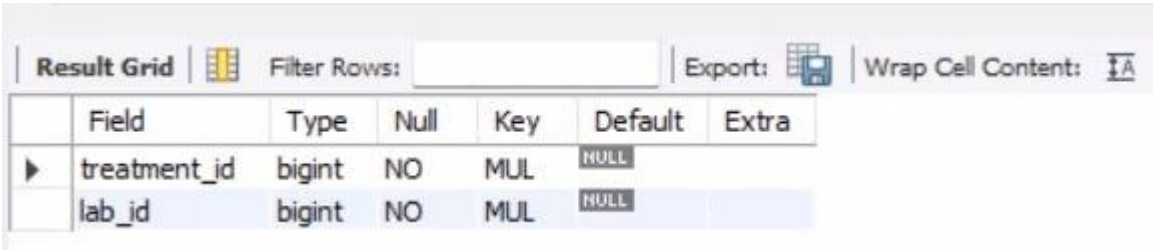
7. **Treatment Table:**



Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
id	bigint	NO	PRI	NULL	auto_increment
medicines	varchar(255)	YES		NULL	
symptoms	varchar(255)	NO		NULL	
treatment_date	date	YES		NULL	
doctor_id	bigint	NO	MUL	NULL	
patient_id	bigint	NO	MUL	NULL	

8. **Treatment_lab Table (link Table):**



Result Grid | Filter Rows: | Export: | Wrap Cell Content:

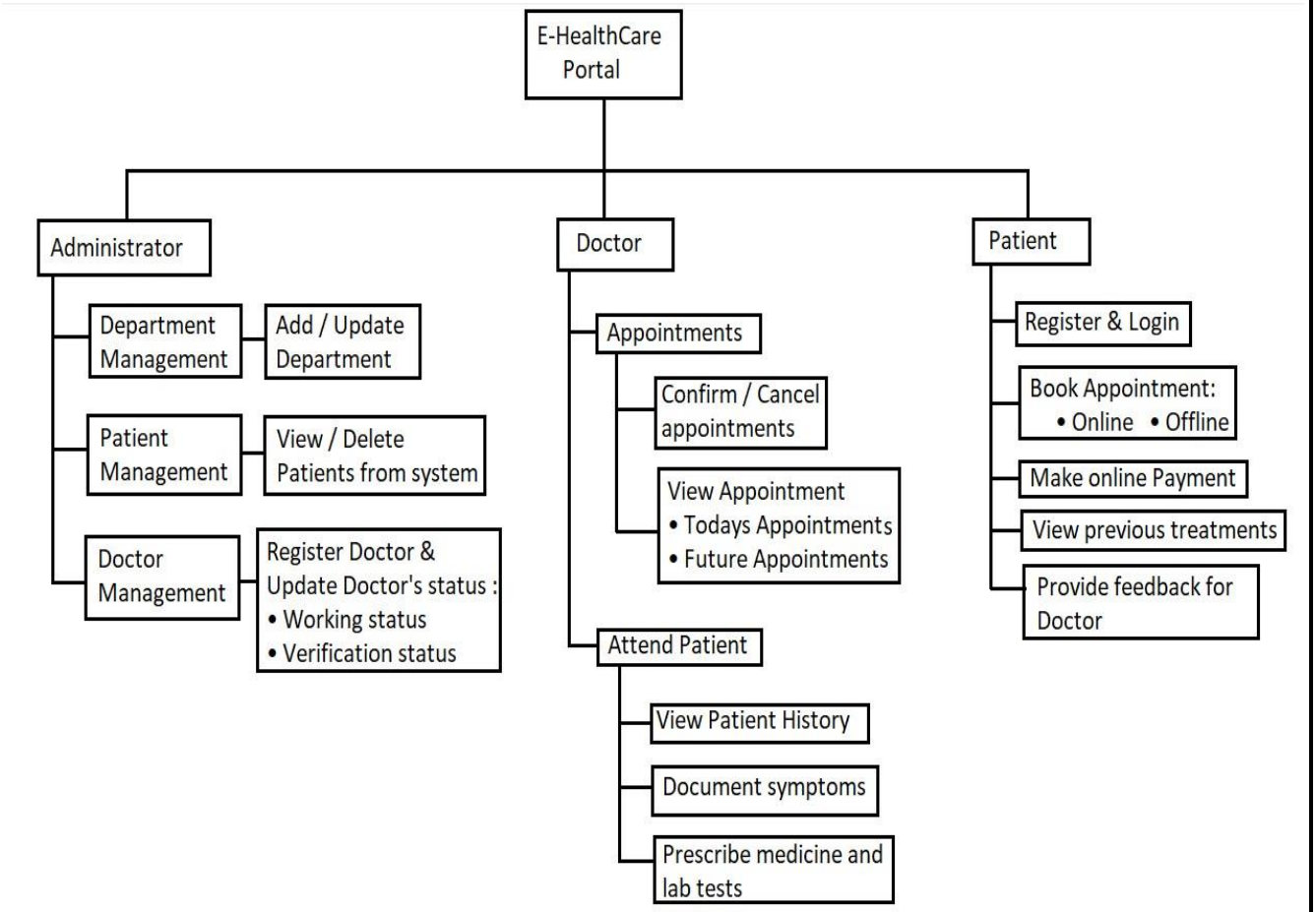
Field	Type	Null	Key	Default	Extra
treatment_id	bigint	NO	MUL	NULL	
lab_id	bigint	NO	MUL	NULL	

9. Appointment Table:

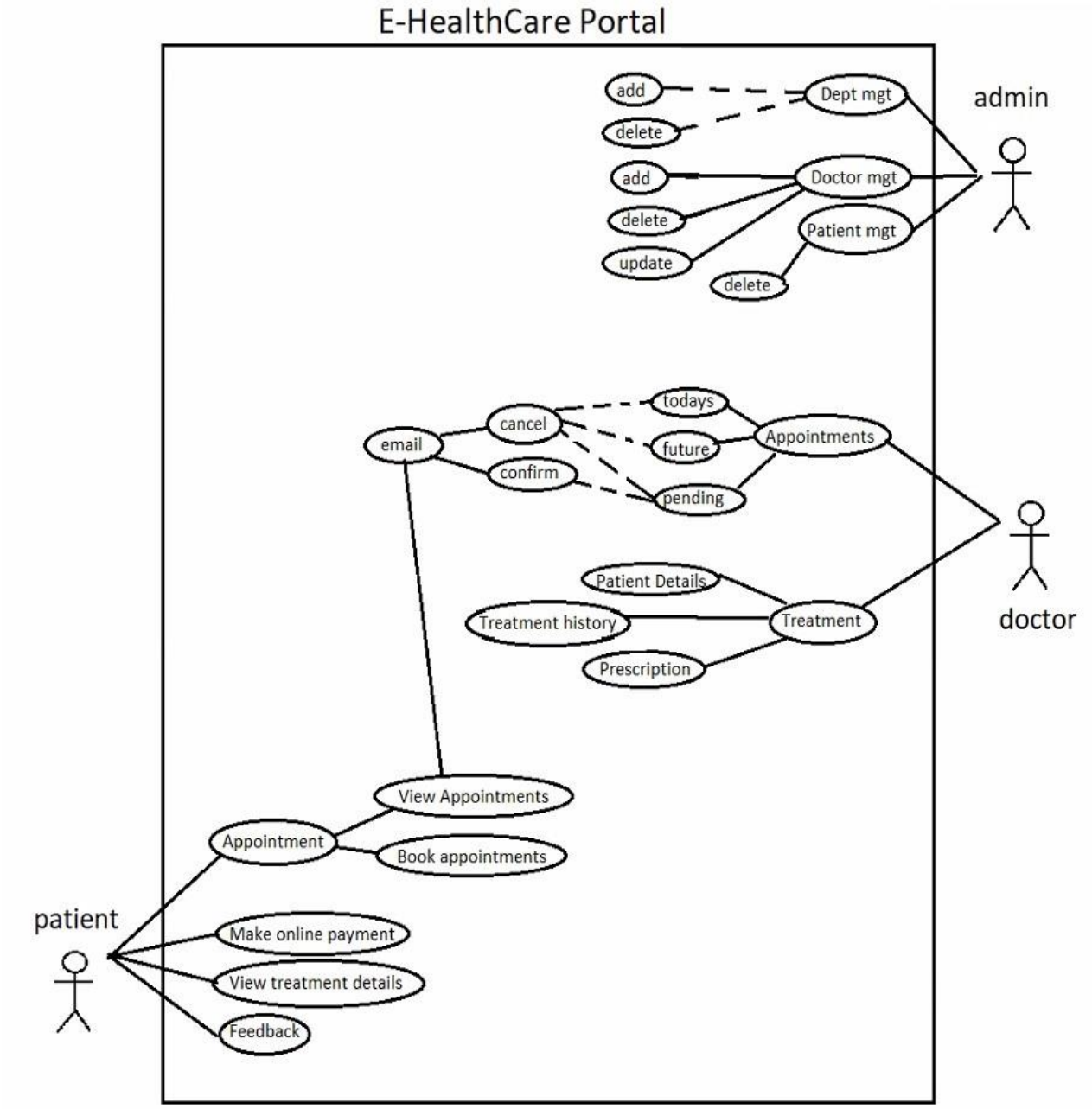
Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
id	bigint	NO	PRI	NULL	auto_increment
date	date	YES		NULL	
online	bit(1)	NO		NULL	
payment_status	bit(1)	NO		NULL	
status	varchar(255)	YES		NULL	
time	time	YES		NULL	
doctor_id	bigint	NO	MUL	NULL	
patient_id	bigint	NO	MUL	NULL	

4.2. FUNCTIONAL DECOMPOSITION DIAGRAM:

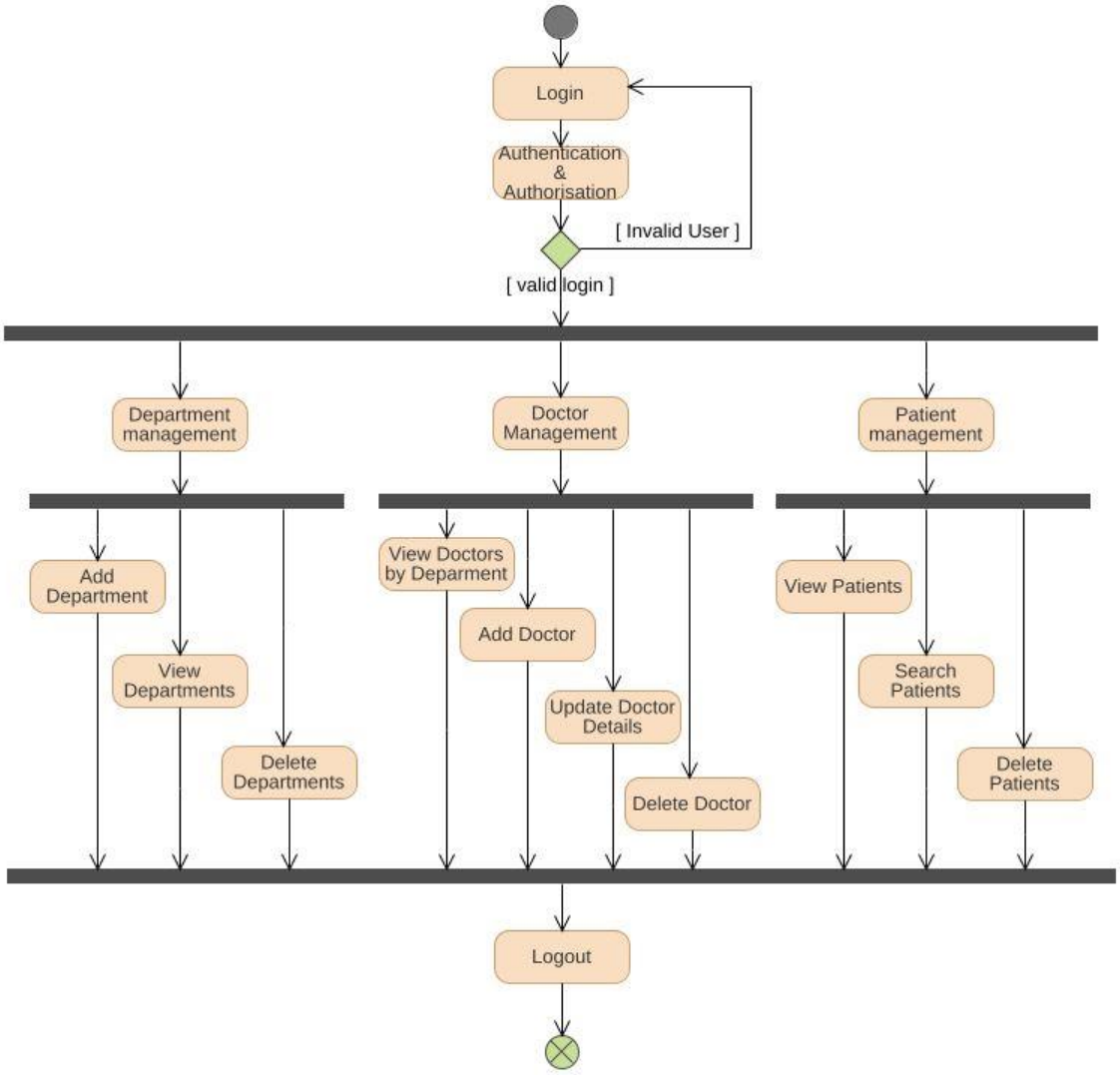


4.3. USECASE DIAGRAM:

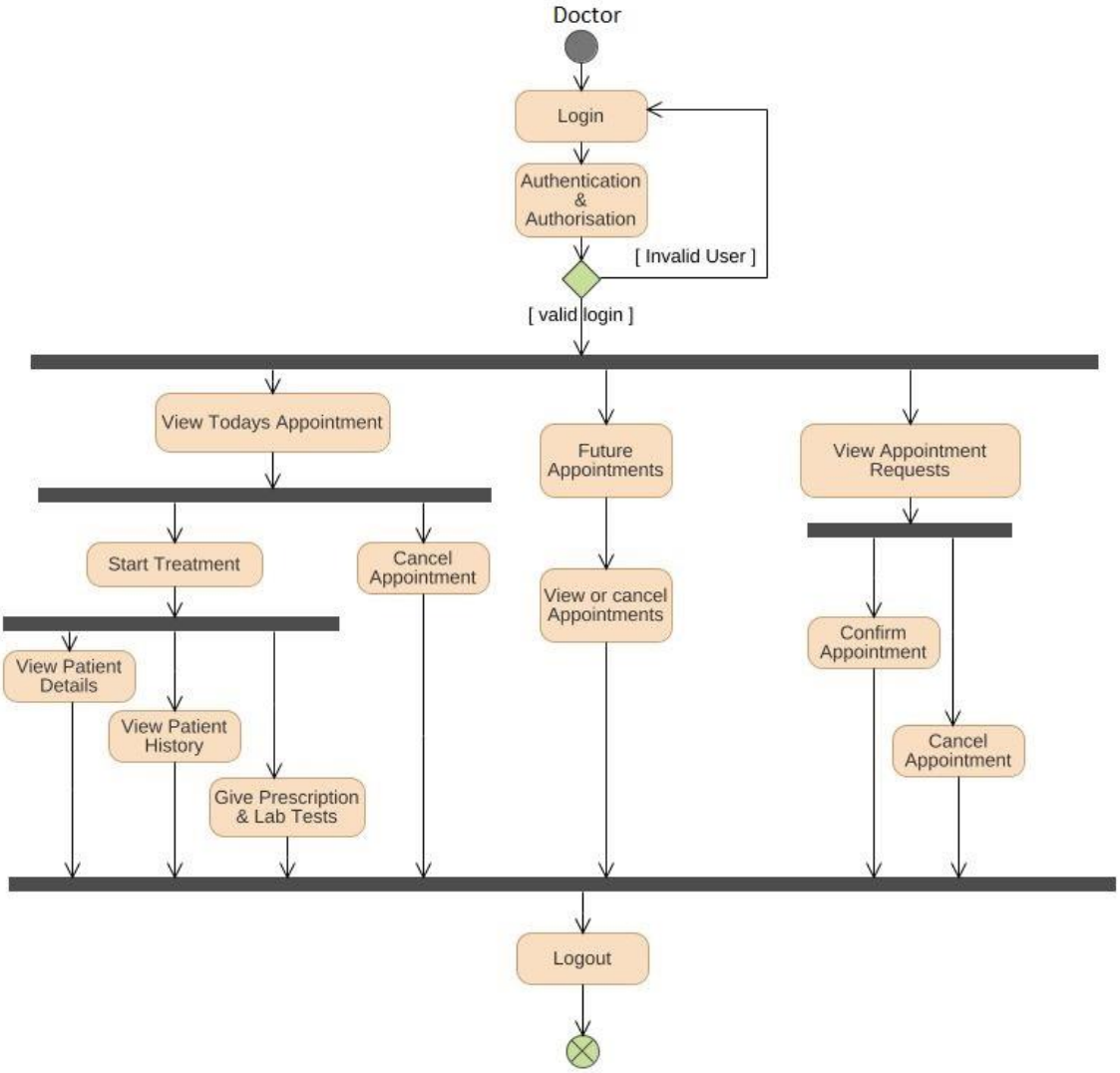


4.4. ACTIVITY DIAGRAM:

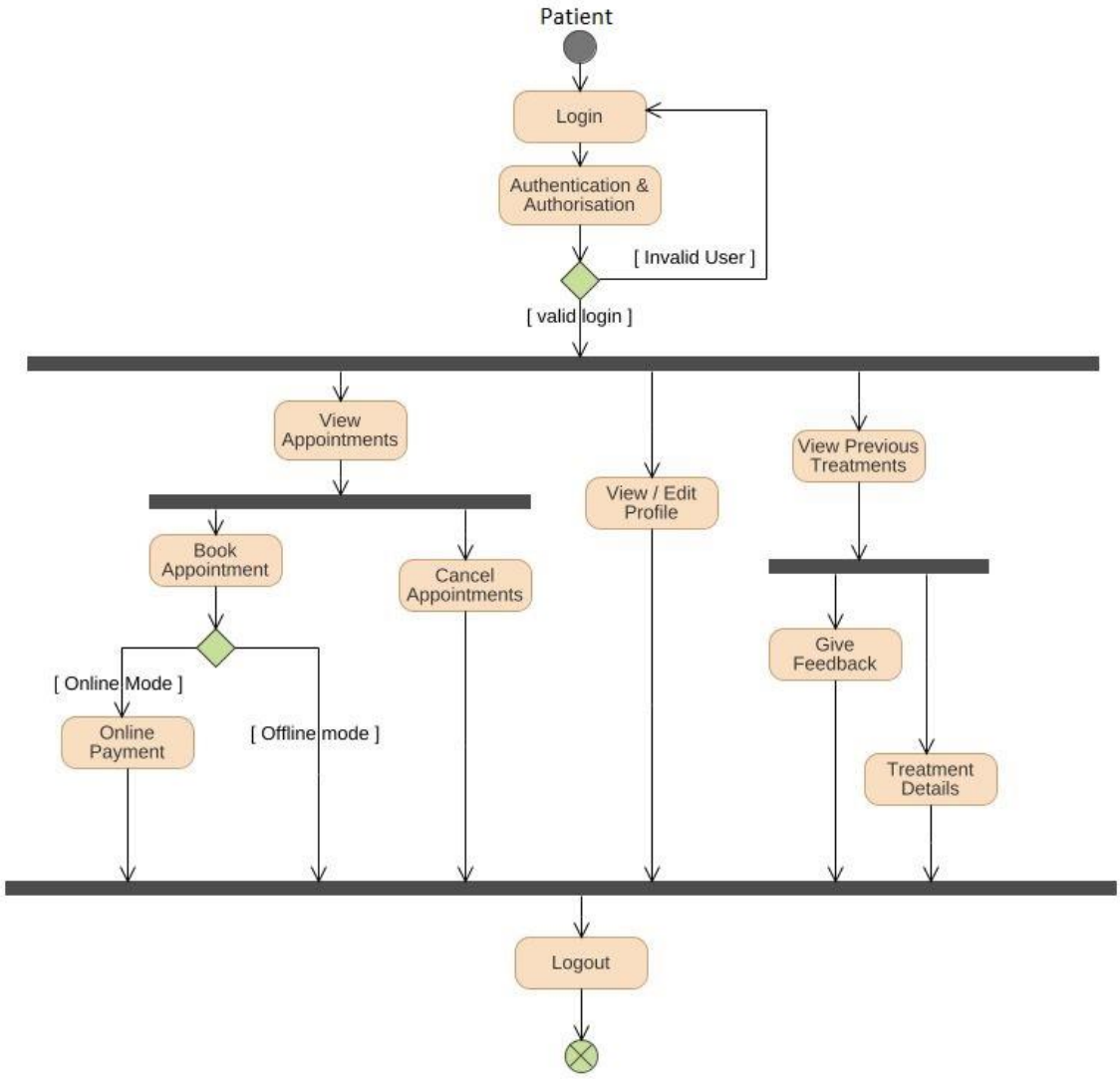
Admin :



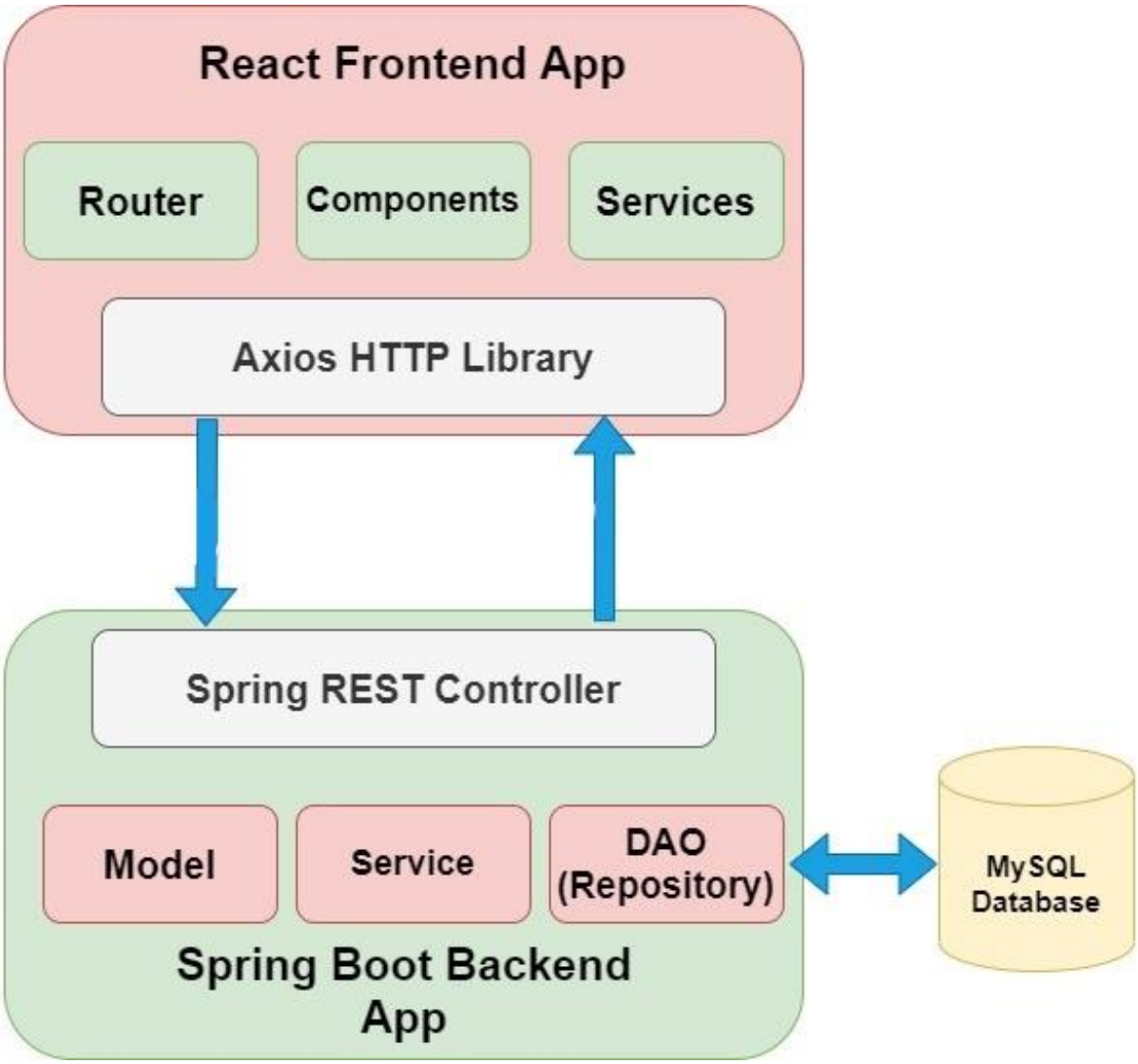
Doctor :


Doctor
Login
Authentication & Authorisation
[Invalid User]
[valid login]
View Todays Appointment
Future Appointments
View Appointment Requests
Start Treatment
Cancel Appointment
View or cancel Appointments
Confirm Appointment
View Patient Details
View Patient History
Give Prescription & Lab Tests
Cancel Appointment
Logout

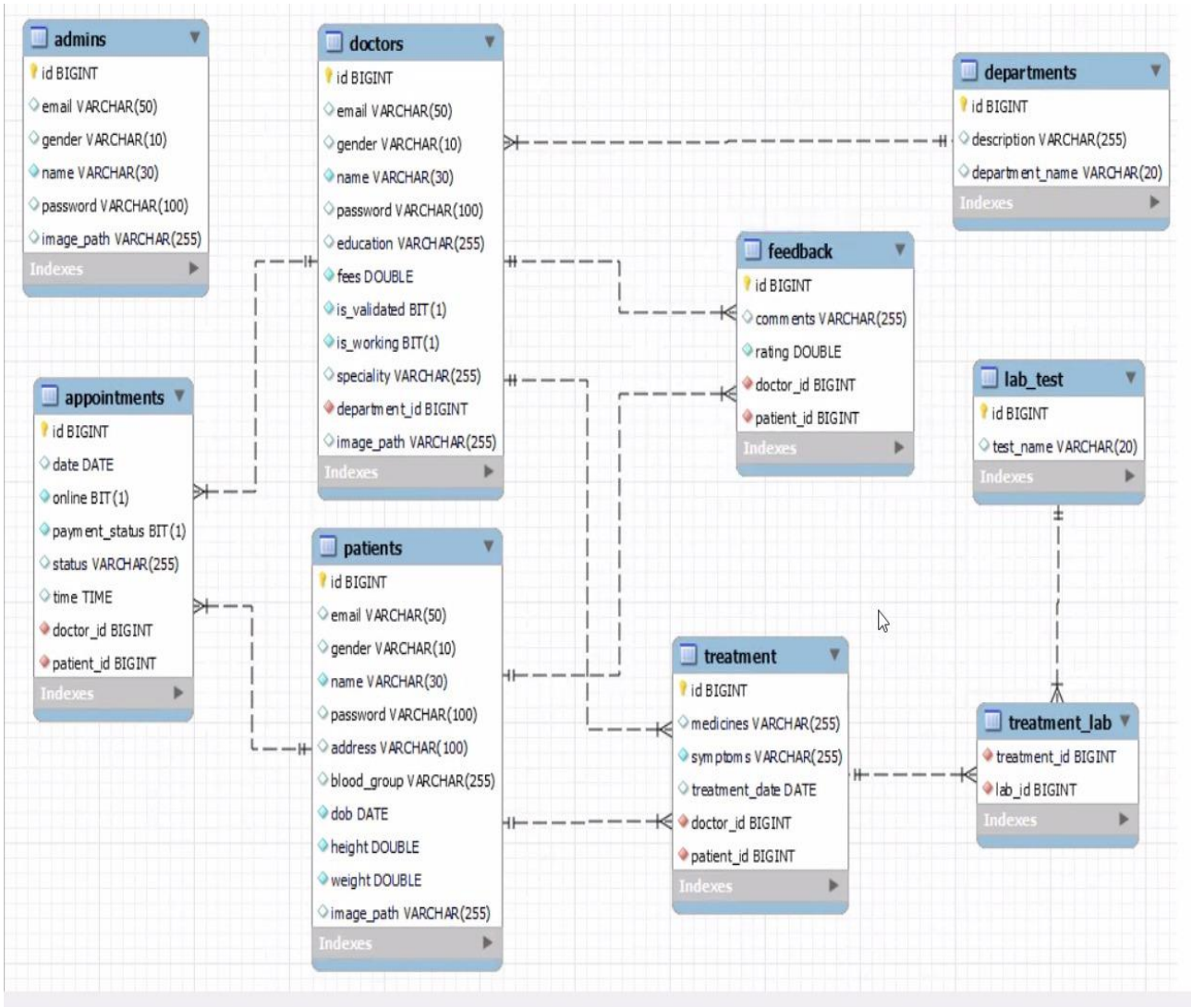
Patient :


Patient
Login
Authentication & Authorisation
[Invalid User]
[valid login]
View Appointments
View / Edit Profile
View Previous Treatments
Book Appointment
Cancel Appointments
Give Feedback
[Online Mode]
Online Payment
[Offline mode]
Treatment Details
Logout

4.5. PROJECT ARCHITECTURE:

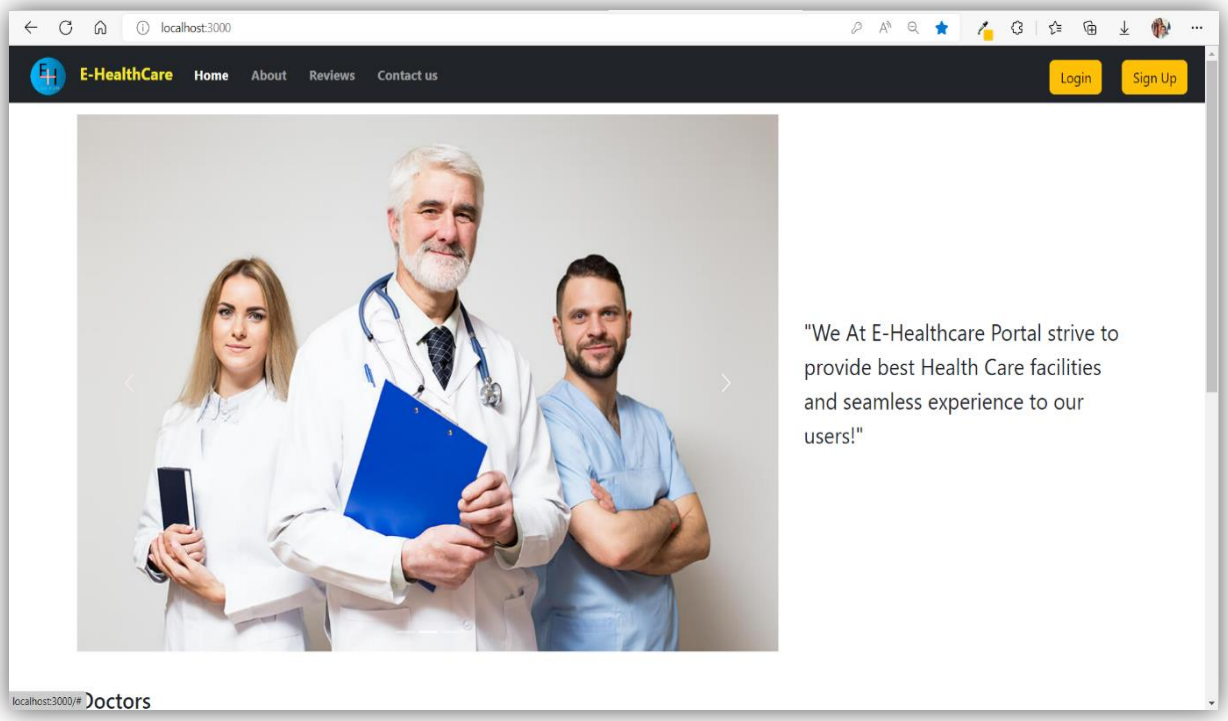


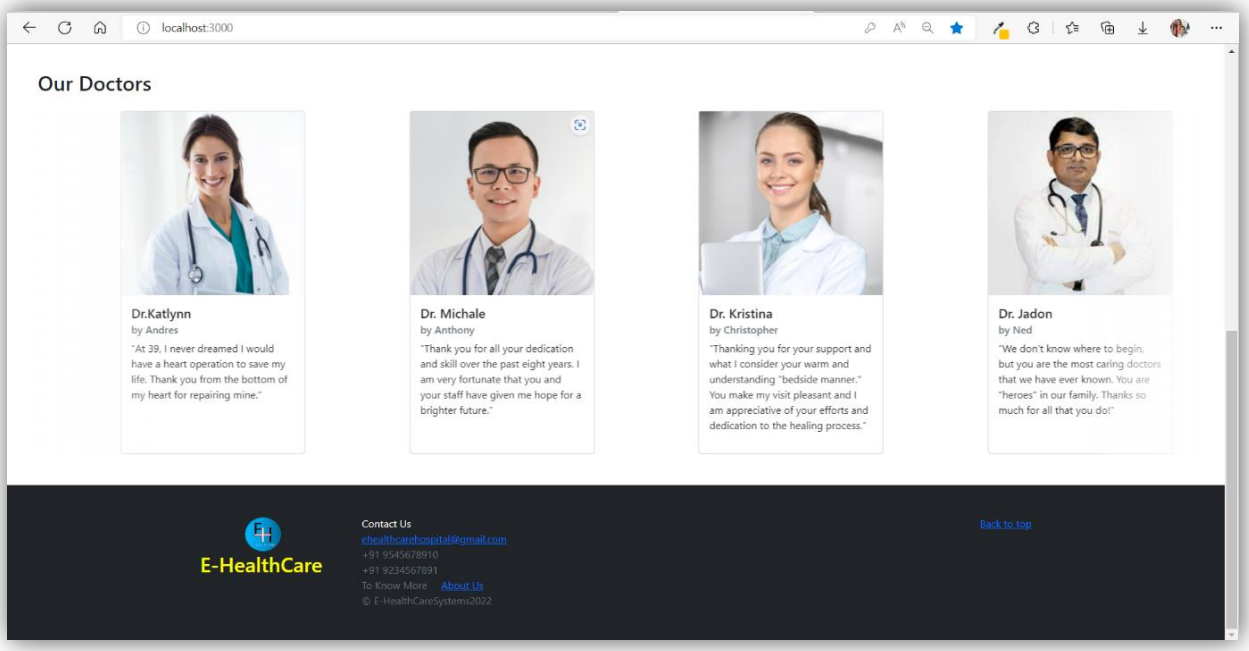
4.6. ER DIAGRAM:



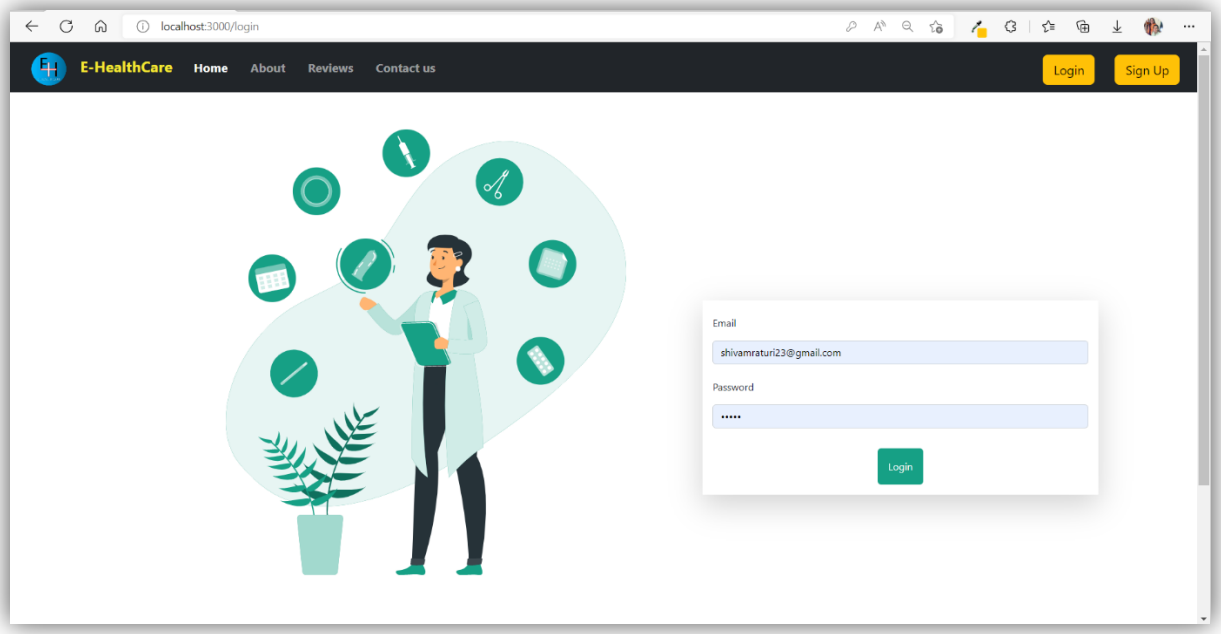
5. PROJECT SCREENSHOTS

1. HomePage-1

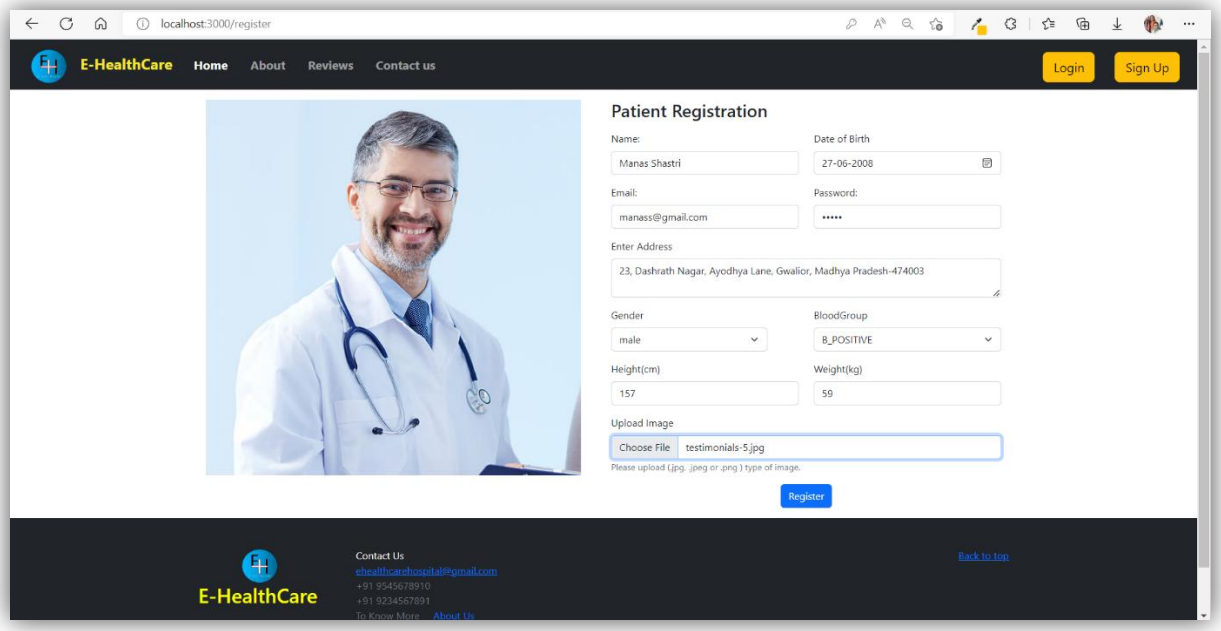




1. Login Page

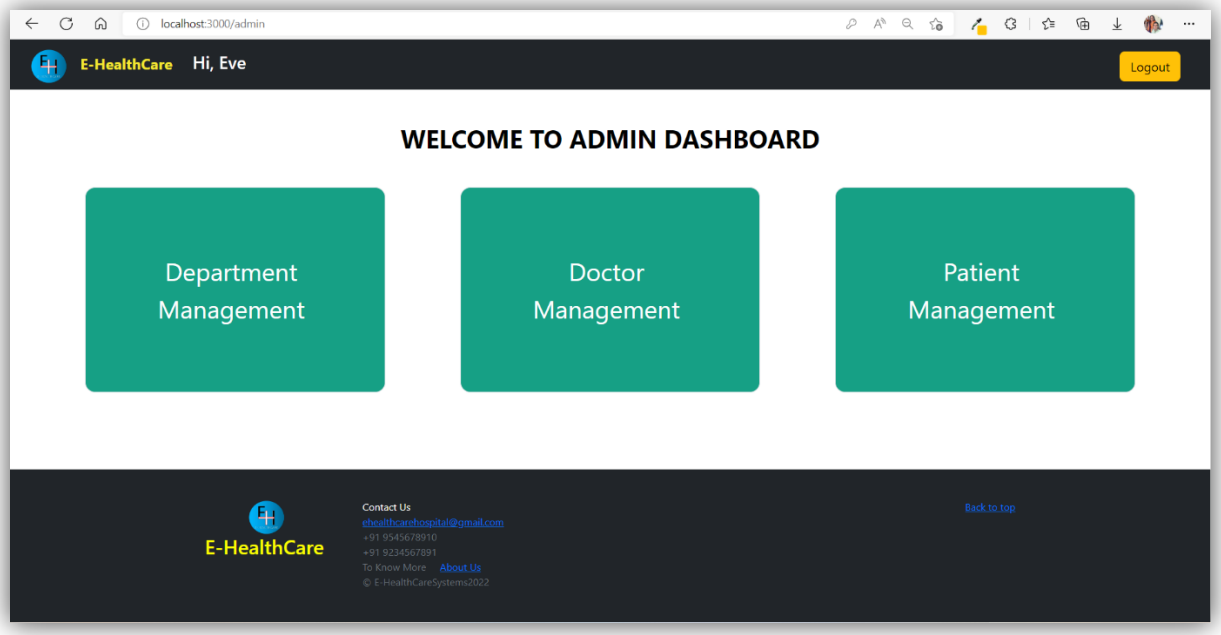


2. Register Page

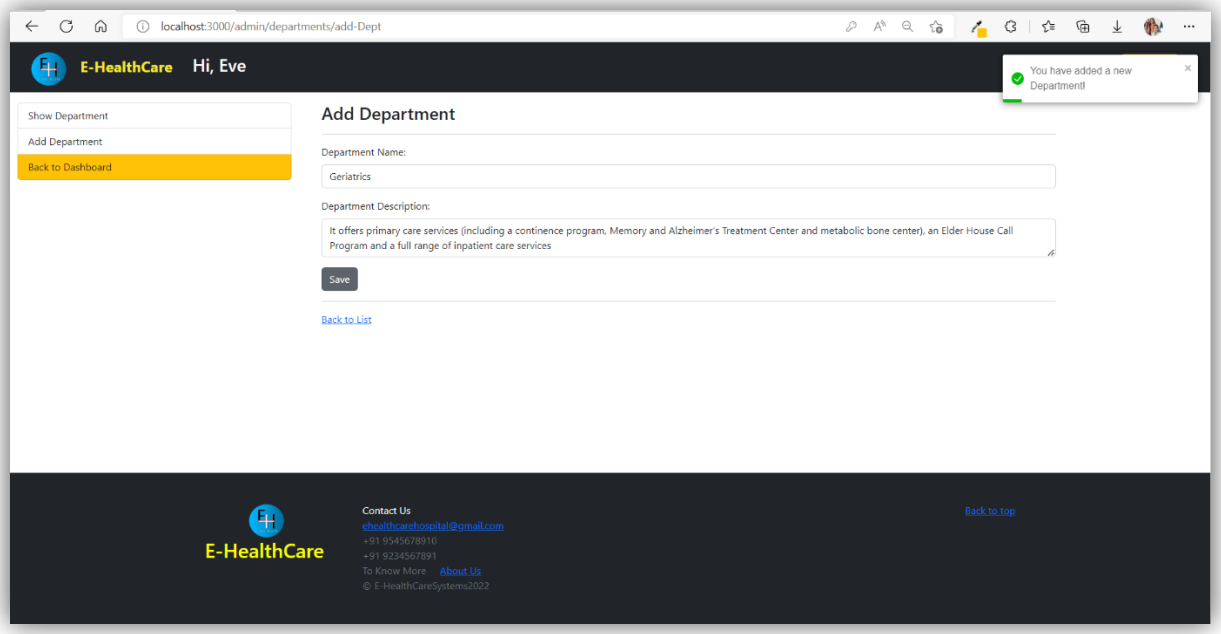


Admin Flow:

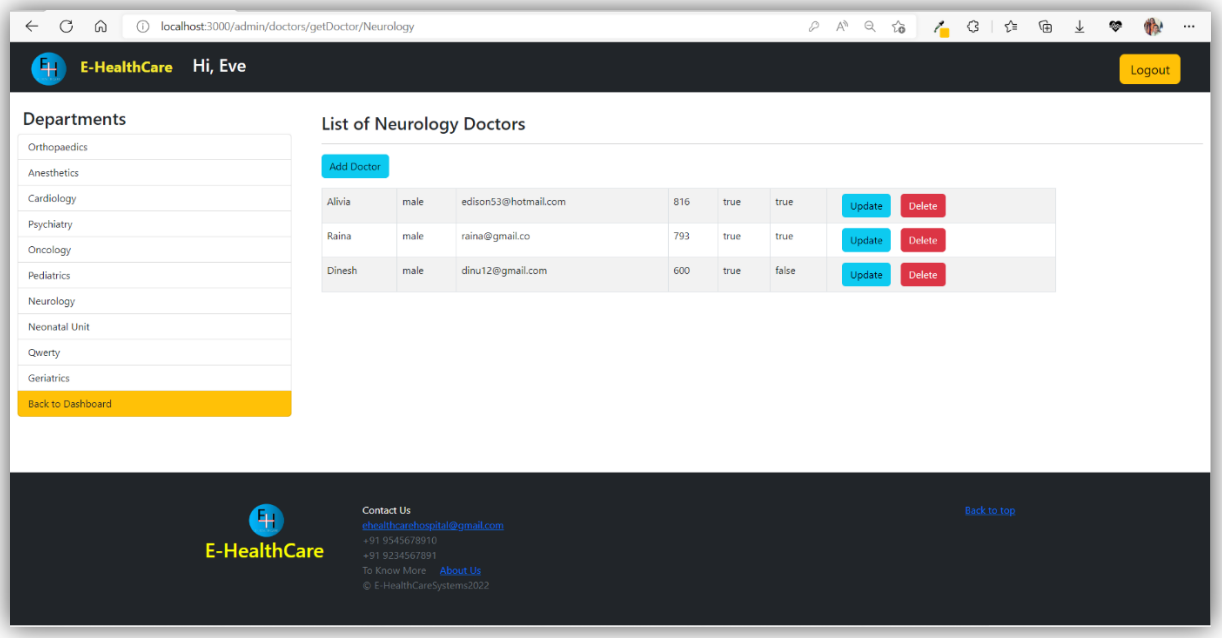
Start Page



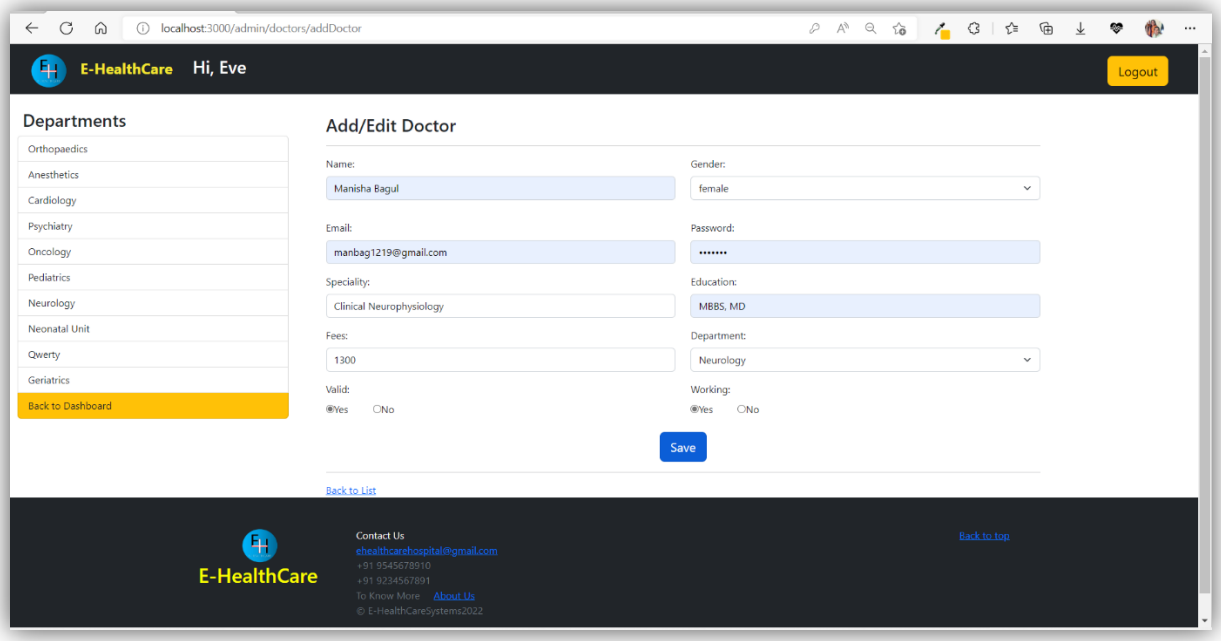
2. **Department Add**



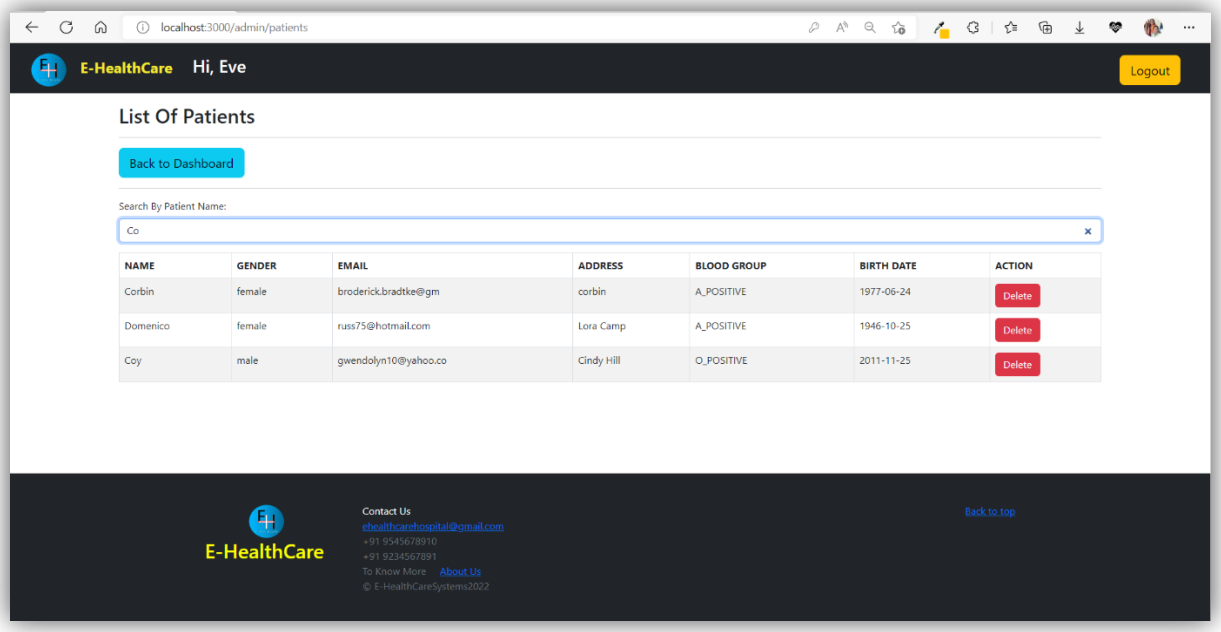
3. Show Doctors from Department



4. Add/Edit Doctors

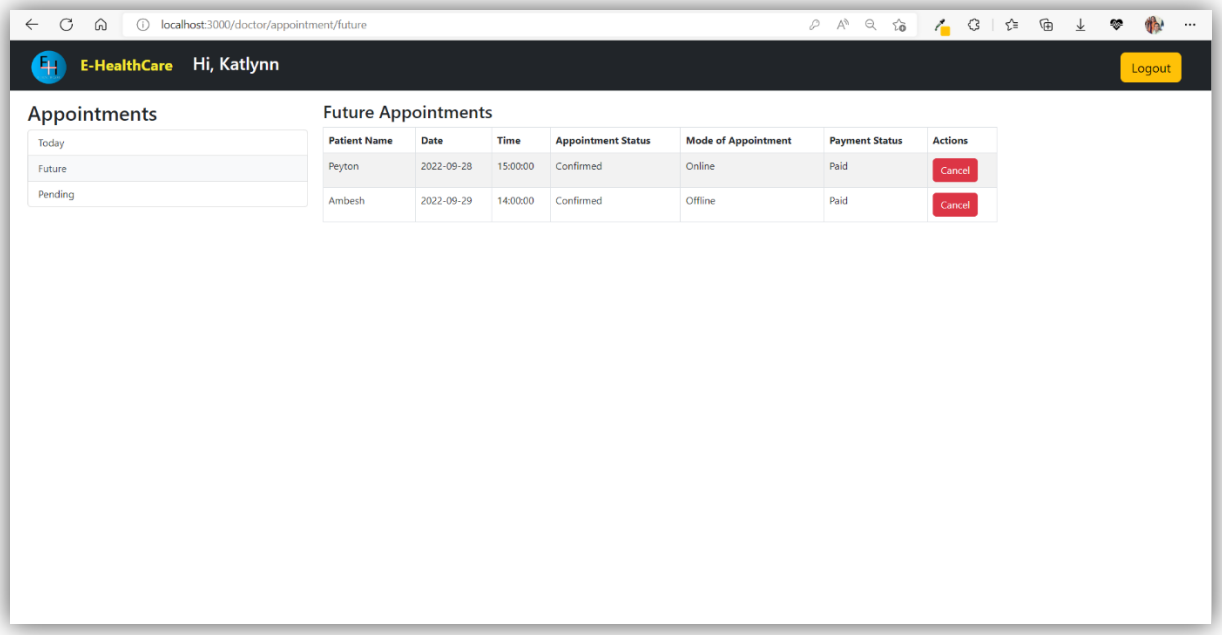


5. Show Patients

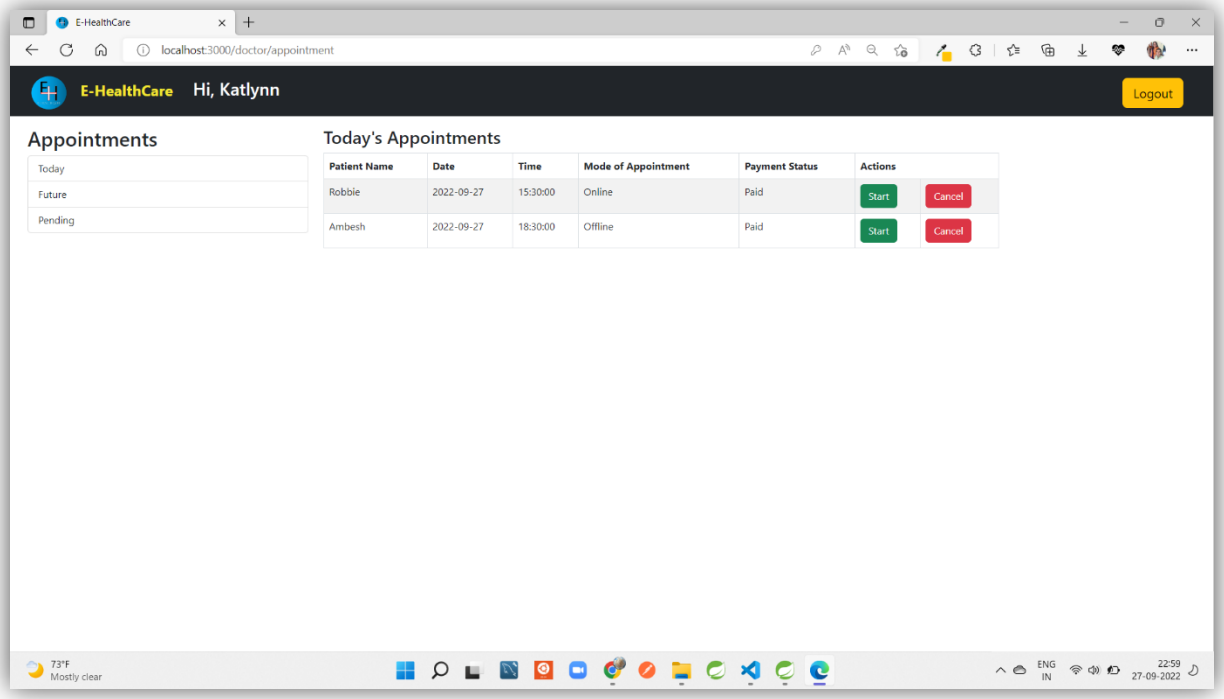


Doctor Flow:

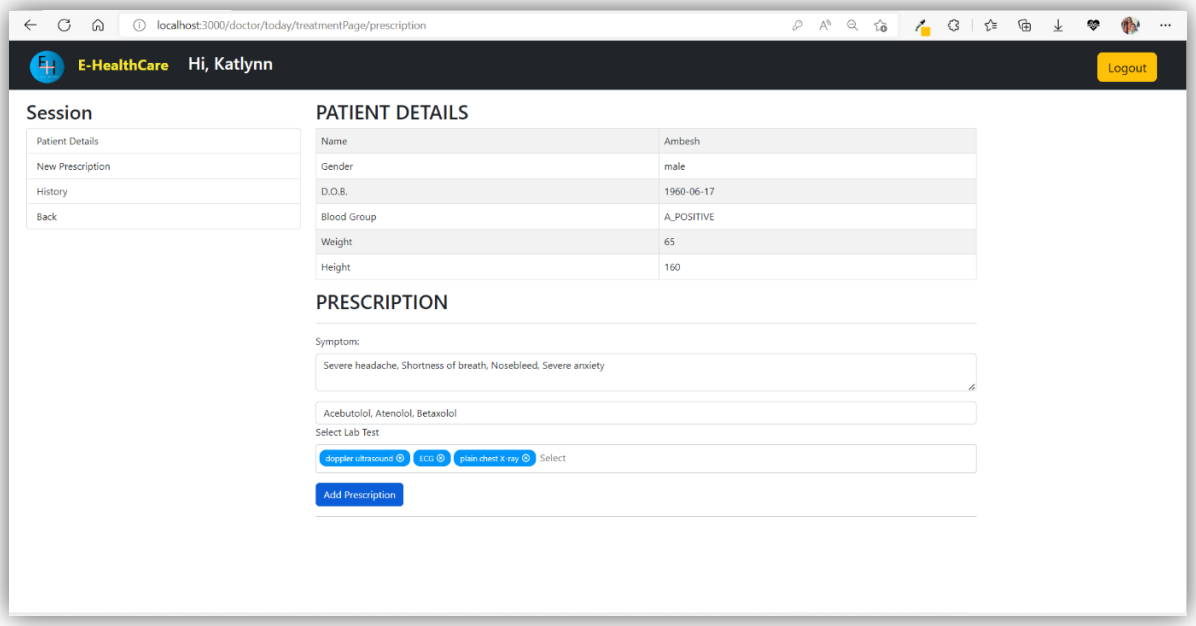
1. Check Appointments



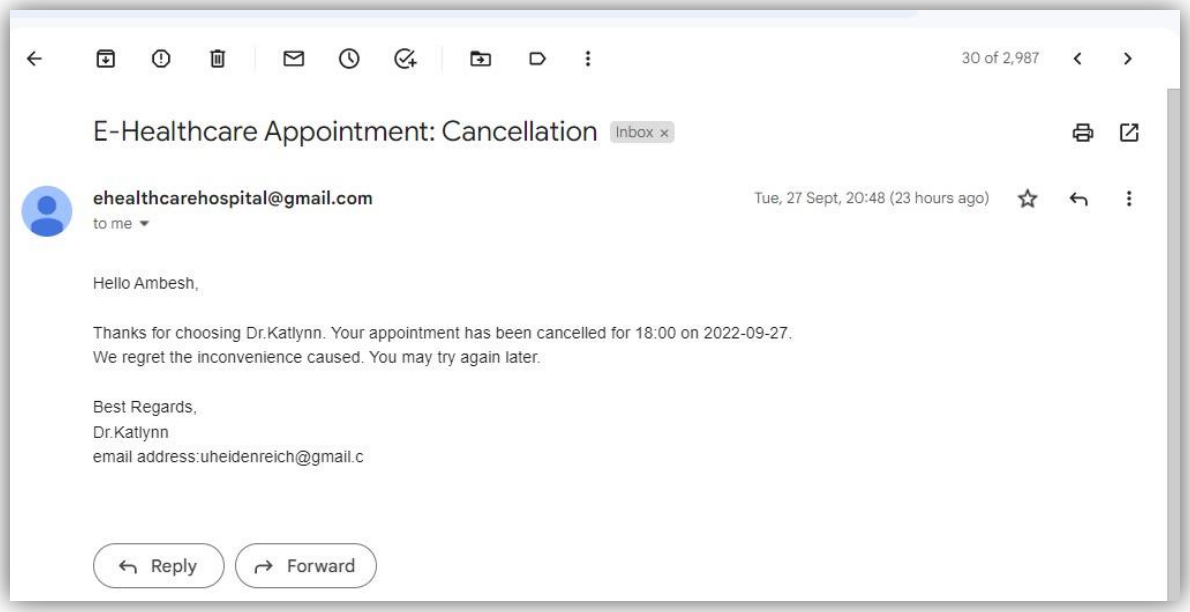
2. Today's Appointments



3. Treatment Details

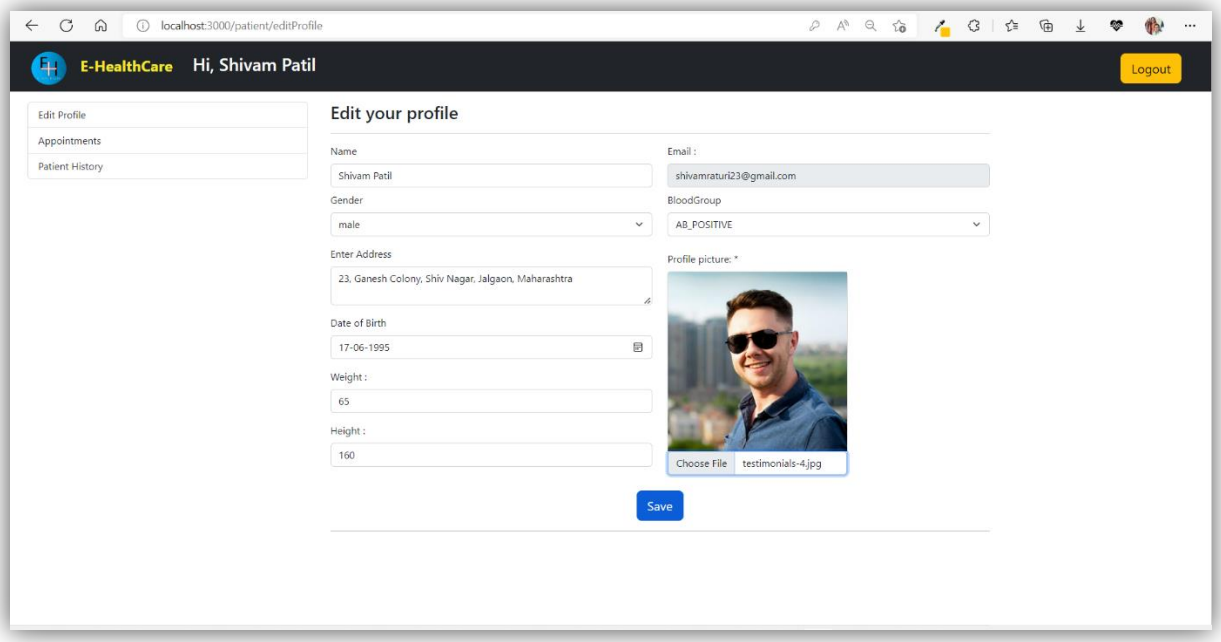


4. Email Sending

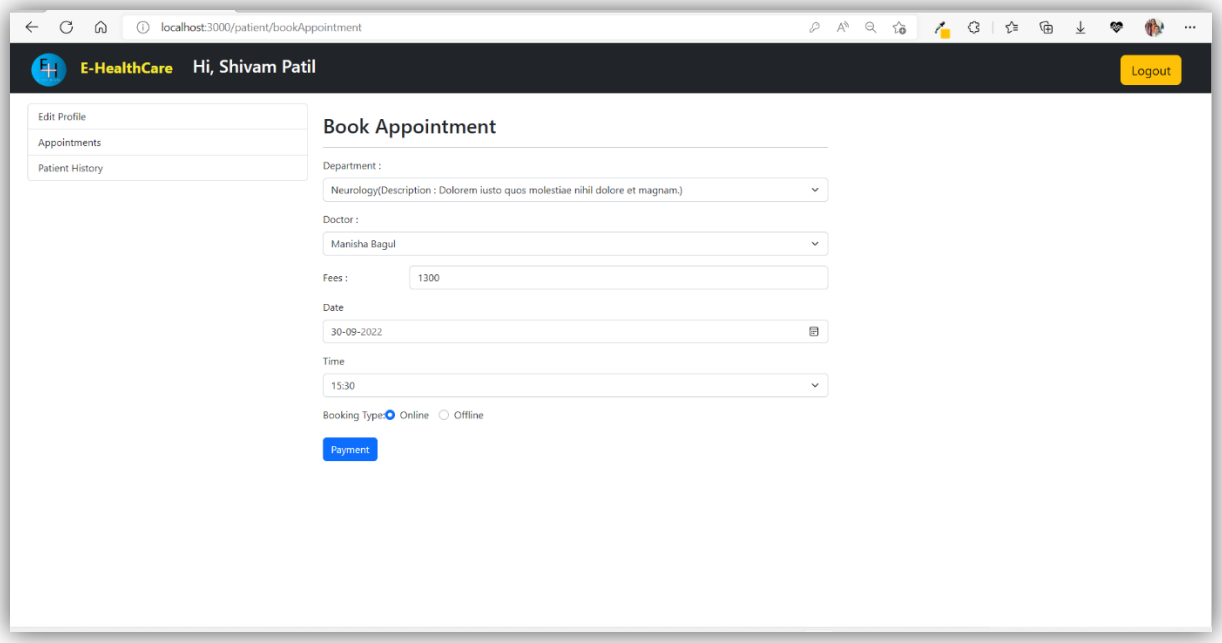


Patient Flow

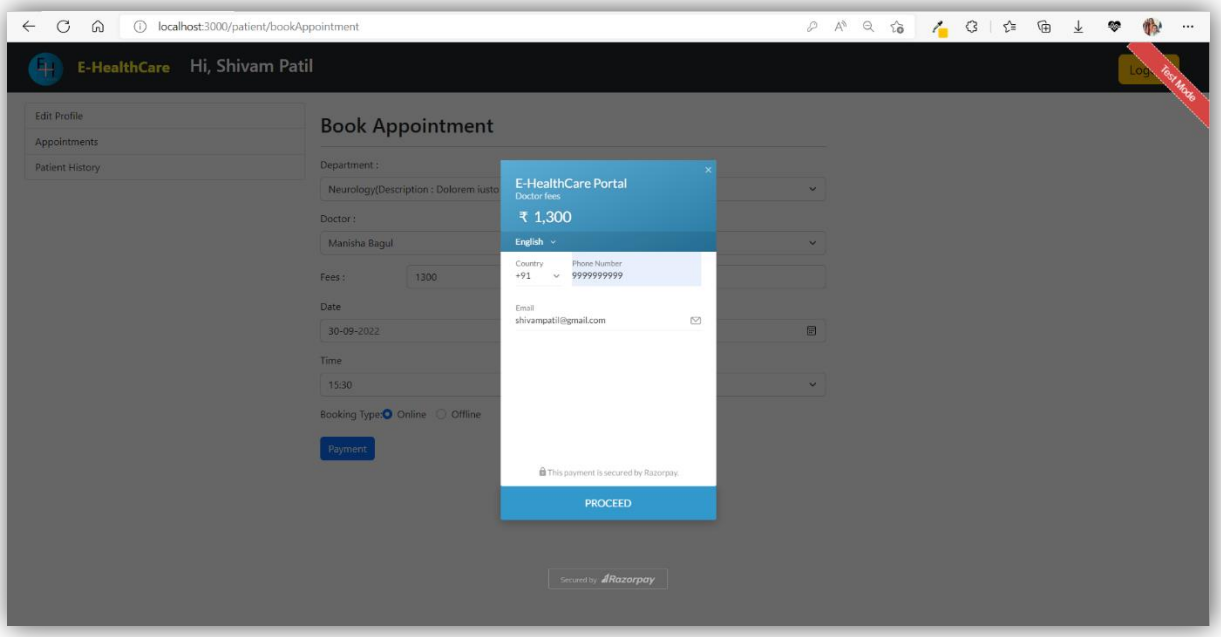
1. Patient Edit Profile

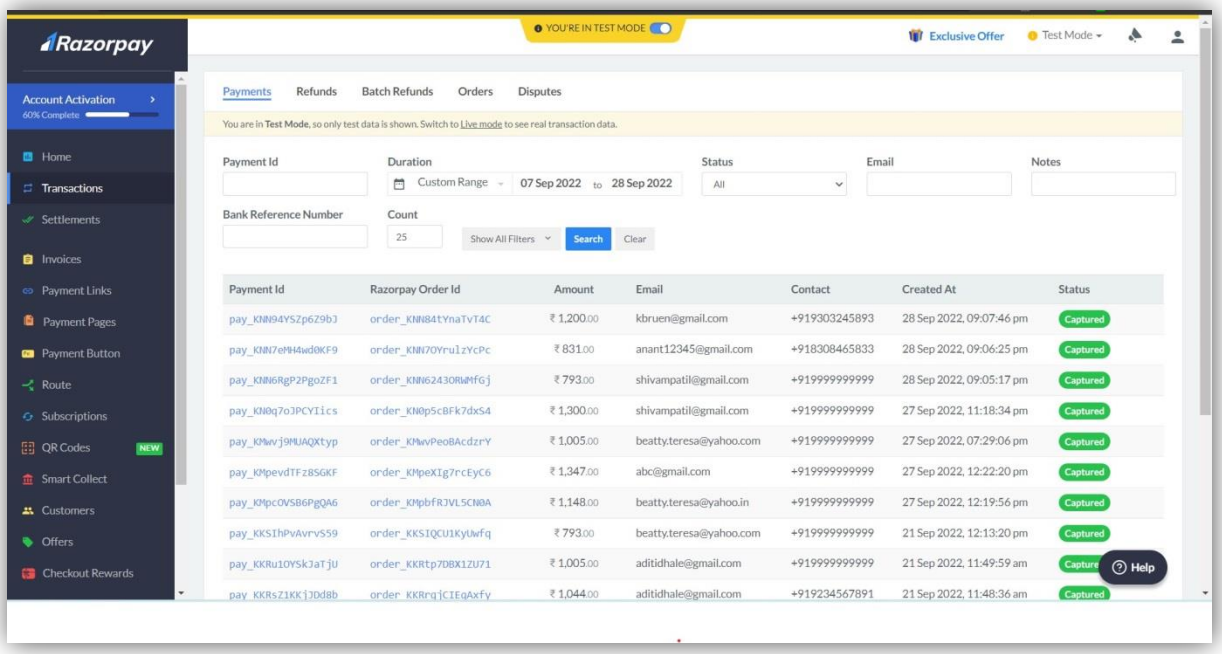


2. Book Appointment

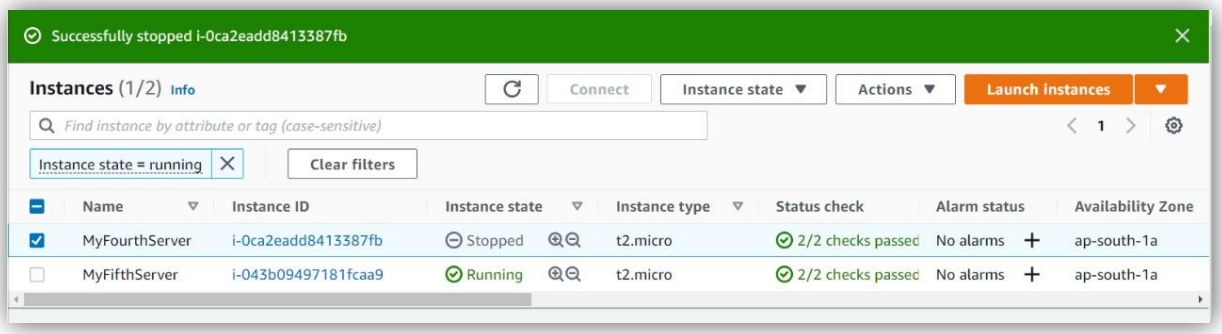


3. Make Payment for Online Appointment





Instance creation on AWS :



Protecting REACT Routes :



6. TESTING

A. ADMIN FEATURE LIST

Sr. No.	Description	Outcome	Result
1.	Login as Admin	Authenticated Admin details are fetched.	Passed
2.	Add Doctor	New Doctor details are added in the database	Passed
3.	Add Department	New Department is added in the database	Passed
4.	Delete Doctor	Associated Doctor details are deleted from the database	Passed
5.	Delete Department	Department is deleted from the database	Passed
6.	Delete Patient	Patient details are deleted from the database	Passed
7.	Update Doctor	Updated doctor details are saved in database	Passed
8.	Get Department List	Department list is fetched from the database	Passed
9.	Get Doctor List by Doctor Name	Got the doctor by searching his name	Passed
10.	Logout	The session was cleared	Passed

B. DOCTOR FEATURE LIST

Sr. No.	Description	Outcome	Result
1.	Login as Doctor	Authenticated Doctor details are fetched.	Passed
2.	Show today Patient List	Today's appointments of patient are fetched from the database	Passed
3.	Show future Patient List	Future's appointments of patient are fetched from the database	Passed
4.	Show Patient's Appointment Request	Pending appointment list of all the patient is fetched.	Passed
5.	Get Patient Details	All the details of particular patient were fetched	Passed
6.	Cancel appointment	Appointment was cancelled by doctor and also its status is updated from "Pending" to "Cancelled" into database	Passed
7.	Add Prescription	All the prescription given by the doctor to patient is saved in database	Passed
8.	Get Patient History	All the records of patient were fetched from the database	Passed
9.	List of Lab Test	All the lab tests entries were fetched from the database	Passed
10.	Logout	The session was cleared	Passed

C. PATIENT FEATURE LIST

Sr. No.	Description	Outcome	Result
1.	Register as Patient	New Patient details saved in the database	Passed
2.	Login as Patient	Fetched authenticated patient details	Passed
3.	Book Appointment	New appointment of a doctor is booked successfully	Passed
4.	Get Patients Appointment	All the appointments booked by patient are fetched	Passed
5.	Cancel Appointment	Appointment is cancelled by patient	Passed
6.	Get Patient Treatment Details	All the treatment details of patient done till now are fetched	Passed
7.	Update Patient Details	Patient updated his profile.	Passed
8.	Feedback of Doctor	The new feedback associated with the doctor was saved in the database	Passed
9.	Online Payment	Online payment is done by patient via Razor Pay	Passed
10.	Logout	The session was cleared	Passed

7. CONCLUSION

"E-HealthCare" an online appointment booking portal, was developed by our project team to simplify the process of booking appointments and their management. We tried using the latest technologies that are cross-platform and robust. Every software we used was open-source in nature, which keeps the cost of production at a minimum.

We were also meticulous about the user experience aspect of our application so that navigating our website is an easy and seamless experience. In conclusion, "E-HealthCare" as an application would be a good choice for any hospital management system that wishes to enter the online domain.

8. FUTURE SCOPE

Using whatever we have learnt over the duration of this course, we tried to make our project as user-friendly and gave it as many features as possible in the limited time allotted for the project work. That said, there are certainly more features that can be added to our application. Some of those are mentioned below:

1. Video Conferencing with Doctor – Portal could include this feature wherein remote patients would be able to consult doctors through video meets.
2. In-house lab test service can be integrated with online payment facility.
3. In case the user forgets the password, a 'reset password' functionality can be added.
4. Native language support - Local crowd can also access the portal in the language they know well.
5. Refund of fees in case the appointment gets cancelled.
6. More staff management features could be added, also inventory management could be included.
7. Web Accessibility can be implemented.
8. After a confirmed purchase, an email with the invoice of the orders can be sent to the customer.

9. REFERENCES

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